ARTIFICIAL INTELLIGENCE · LECTURE 06

Searching With a Hint

*Informed Search: Heuristics, Greedy Best-First & A**



CLO-2 · Apply

AIMA (3rd ed.), Chapter 3

Muhammad Waqas Ishaq, Instructor

WHERE WE'RE HEADED

Learning Objectives

By the end of this lecture you will be able to:

1. Explain the difference between blind search and informed search.
2. Define a heuristic h(n) in plain words, with examples you can compute.
3. Trace Greedy Best-First Search step by step, and see where it gets fooled.
4. Calculate f(n) = g(n) + h(n) and trace A* on a road map.
5. Explain “admissible” and “consistent” hints in simple words.
6. Compare Greedy and A*, and explain how the hint changes the search.

Russell & Norvig, Artificial Intelligence: A Modern Approach (3rd ed.), Chapter 3, Section 3.5

LAST TIME → TODAY

Last Time We Searched Blind. Today We Get a Hint.

Same search recipe — one extra piece of information changes everything.

Lecture 04 — Uninformed Search

The agent knew the map, available actions and road costs, but had NO heuristic estimate of how far each city was from the goal. So it explored by a fixed rule: level by level (BFS), deepest first (DFS), or lowest cost so far (UCS).

Real life: Looking for your friend in a huge mall by checking every shop, one by one.

Today — Search With a Hint

We give the agent a hint: an ESTIMATE of how far each place is from the goal. Now it can head in the promising direction instead of exploring everywhere.

Real life: Your friend texts “I’m near the food court.” Now you walk toward the food court — you skip most shops.

Informed search = the same search recipe + a hint. Today: heuristics, Greedy Best-First Search and A*.

Russell & Norvig, Artificial Intelligence: A Modern Approach (3rd ed.), Chapter 3, Section 3.5

THE KEY WORD · AIMA SECTION 3.5

What Is a Heuristic?

A heuristic is an educated guess — cheap to calculate, usually helpful, not always exact.

THE DEFINITION

h(n) = estimated cost from node n to the nearest goal

Read it as: "From here, how far do I THINK the goal is?"

3 THINGS TO REMEMBER

h(n) ≥ 0 a distance cannot be negative

h(goal) = 0 at the goal, nothing is left to travel

It is a GUESS it need not be perfect

HEURISTICS YOU USE EVERY DAY

Taxi driver

"Airport? About 30 minutes." A guess made before the trip starts.

Cricket chase

Required run rate: a quick estimate of how hard the chase is.

Choosing a queue

You pick the shortest-looking line — you don't count every person.

Heuristics You Can Compute Yourself

Both are quick to calculate — no searching needed to get the estimate.

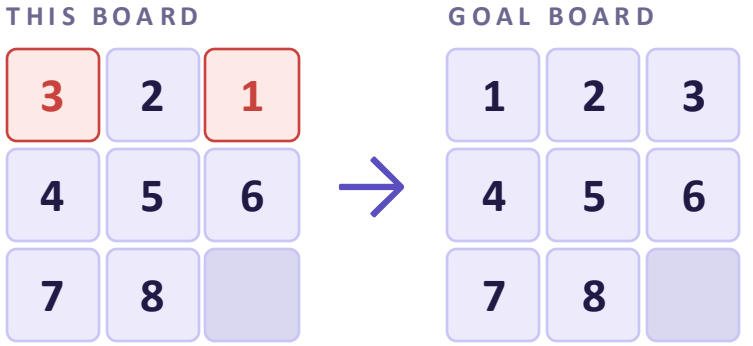
Route-Finding

Hint: straight-line distance to the goal

"As the crow flies" to Islamabad. It is easy to get from a map or GPS coordinates — and it is never longer than the real road, because a road can bend but cannot be shorter than a straight line.

Why it helps: a city 100 km away "as the crow flies" is a better bet than one 200 km away — so we try it first.

The 8-Puzzle



Hint 1 — Misplaced tiles: count tiles not at home. Tiles 3 and 1 are wrong → $h_1 = 2$.

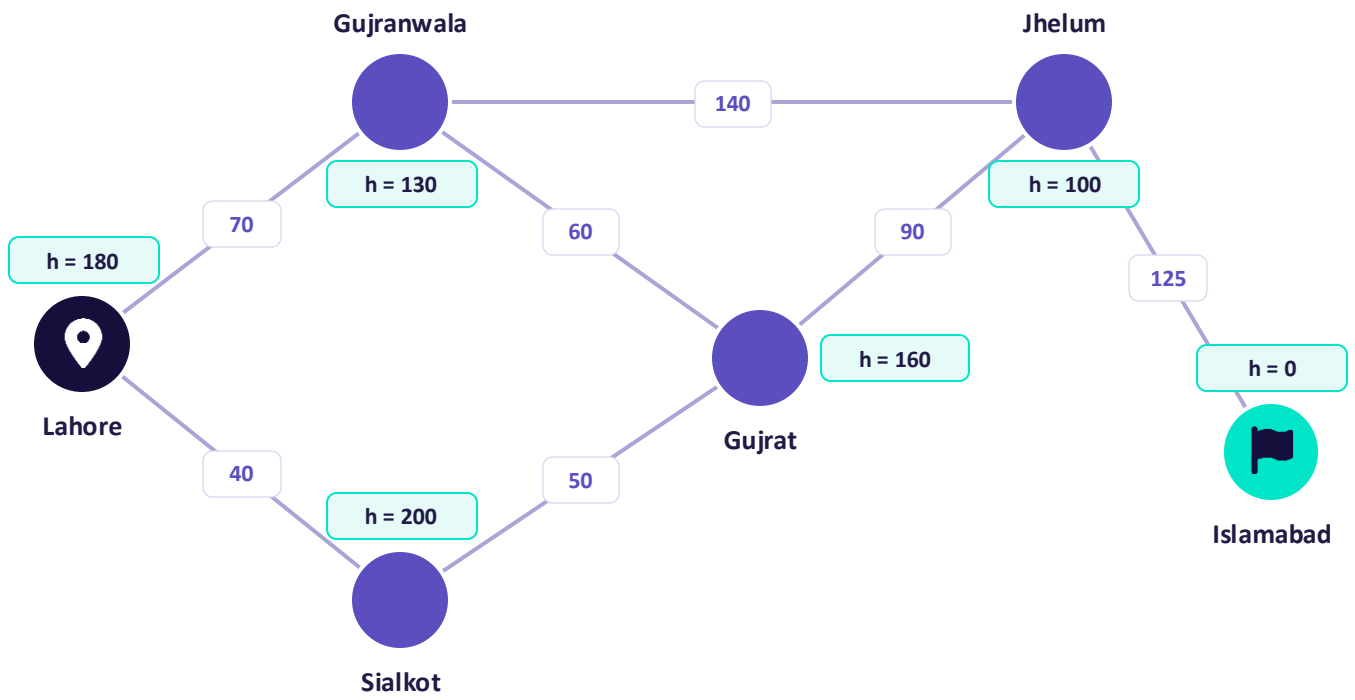
Hint 2 — Manhattan distance: add up how many squares each tile must slide. Tiles 3 and 1 are each 2 squares from home →

Manhattan is the sharper hint: it measures HOW FAR, not just "wrong or right".

OUR MAP FOR TODAY

Lahore → Islamabad, Now With Hints

The same style of map as Lecture 04 — but every city now carries a hint.



HOW TO READ IT

Number on a road
the REAL distance in km. We pay this.

Green tag h = …
our GUESS of the straight-line km from that city to Islamabad.

Notice
Jhelum looks very close (h = 100). Sialkot looks far (h = 200).

All road costs and h-values are illustrative teaching numbers, not actual geographic measurements.

ALGORITHM · AIMA SECTION 3.5.1

Greedy Best-First Search

Always go to the place that LOOKS closest to the goal.

THE IDEA

Of all the places you could explore next, pick the one with the SMALLEST h(n) — the one that looks closest to the goal.

REAL LIFE

Walking toward a tall tower in a new city: at every junction you take the road that points most directly at the tower.

HOW IT IS BUILT

The frontier is a PRIORITY QUEUE ordered by h(n). The smallest h is always served first.

✓ Fast — heads for the goal

✕ Ignores cost already paid

THE EVALUATION FUNCTION

$$f(n) = h(n)$$

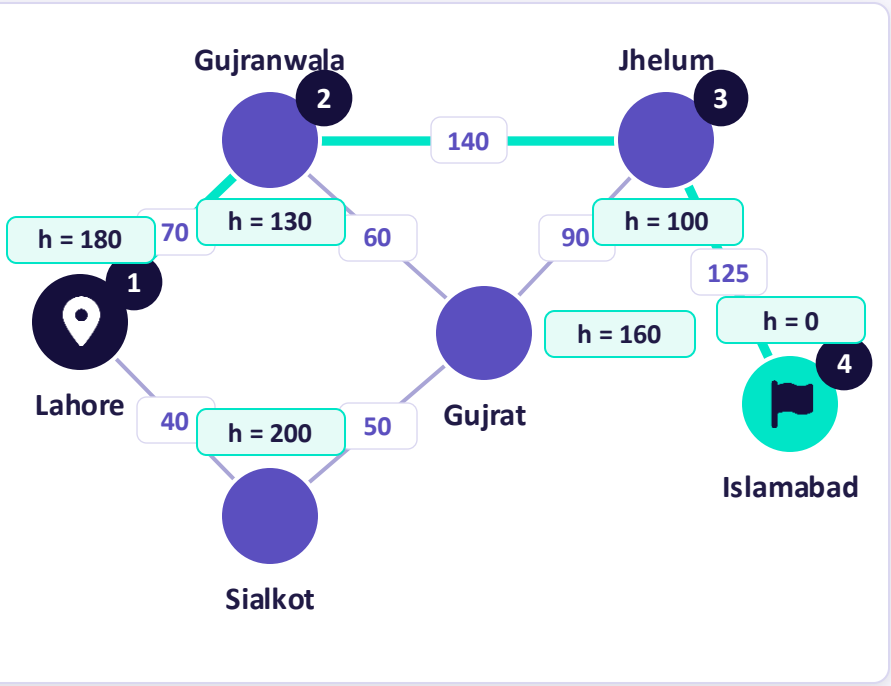
Score = hint only. Lowest score goes first.

THE RECIPE

1. Put the start city in the frontier.
2. Take the city with the smallest h.
3. Is it the goal? Yes → stop. No → continue.
4. Add its unexplored neighbours to the frontier.
5. Repeat from step 2.

Greedy Trace: Lahore → Islamabad

The frontier is a priority queue — the SMALLEST h in the whole frontier is always taken first.



Step	Take (smallest h)	h	Goal?	Frontier now (city: h)
1	Lahore	180	No	Gujranwala: 130, Sialkot: 200
2	Gujranwala	130	No	Jhelum: 100, Gujrat: 160, Sialkot: 200
3	Jhelum	100	No	Islamabad: 0, Gujrat: 160, Sialkot: 200
4	**Islamabad**	**0**	**YES — goal!**	—

PRIORITY QUEUE = emergency room. The case that looks most urgent (smallest h) is seen first.

Step 1. Lahore is taken. Its neighbours Gujranwala (h=130) and Sialkot (h=200) join.

Step 2. Gujranwala has the smallest h, so it goes next. It adds Jhelum (100) and Gujrat (160).

Step 3. Jhelum (h=100) is now the smallest. It adds Islamabad (h=0).

Step 4. Islamabad has h=0 — the goal. Stop.

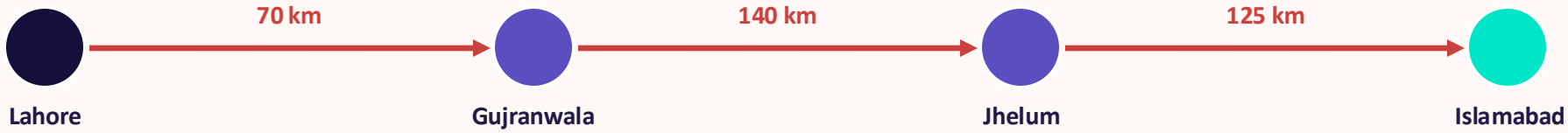
Result: Lahore → Gujranwala → Jhelum → Islamabad = 70 + 140 + 125 = 335 km. Only 4 cities checked.

THE CATCH

Greedy's Trap: Fast, But Not Always Cheapest

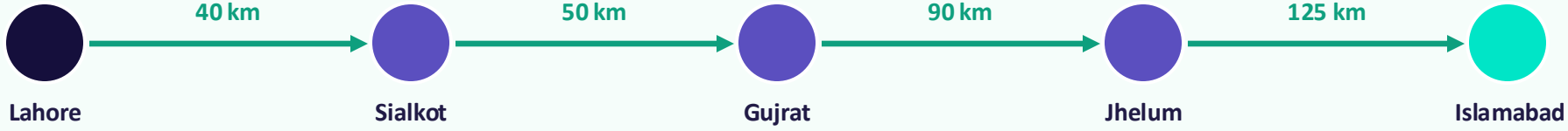
Same map, two different routes. Which one would you rather pay for?

GREEDY FOUND THIS · 4 CITIES CHECKED



335 km
total cost

THE CHEAPEST ROUTE · EXISTS ON THE SAME MAP



305 km
total cost

Why? Jhelum looked great (h = 100), but the 140 km road to reach it was expensive. Greedy never looks at the cost already paid.

Fix coming up in this lecture: A* adds the cost already paid to the hint.

Three Numbers: g, h and f

A gives every city a score. The lowest score is served first.*

g(n) Cost so far

The REAL cost already paid to reach this city from the start.

Taxi: the meter reading, Rs 400.

h(n) Estimated cost to go

The hint: our GUESS of the cost still left to reach the goal.

Taxi: the driver says "about Rs 300 more".

f(n) Estimated total

g + h: our best guess of the whole trip if we go through this city.

Taxi: Rs 400 + Rs 300 = about Rs 700.

A* EVALUATION FUNCTION

$$f(n) = g(n) + h(n)$$

Compare: UCS uses g only. Greedy uses h only. A uses both.*

On our map: Gujranwala has g = 70 (road from Lahore) and h = 130 (hint). So f = 70 + 130 = 200.

ALGORITHM · AIMA SECTION 3.5.2

A* Search

Always continue from the city with the lowest f = g + h.

THE IDEA

Pick the city whose total estimated trip (cost so far + guess to go) is smallest.

REAL LIFE

Choosing a route by checking the meter AND the driver's estimate, not just how close the destination looks.

HOW IT IS BUILT

The frontier is a PRIORITY QUEUE ordered by f(n). Like UCS — but with the hint added.

✓ Cheapest route (good hint)

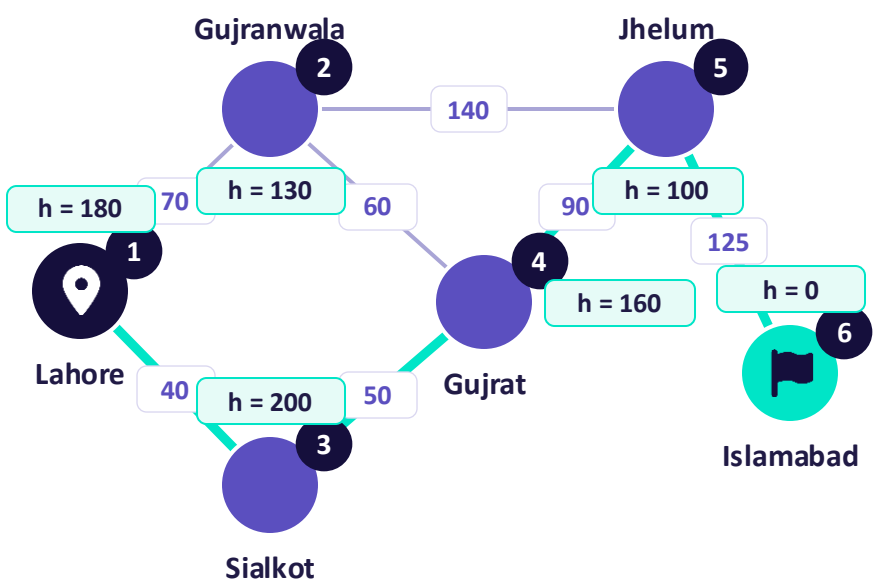
✕ Needs a lot of memory

THE RECIPE

1. Put the start city in the frontier (g = 0).
2. Take the city with the LOWEST f = g + h.
3. Is it the goal? Yes → stop. No → continue.
4. For each neighbour: g = g(city) + road cost; then f = g + h.
5. New city → add it. Already in the frontier with a higher g? → UPDATE it to the cheaper one.
6. Repeat from step 2.

A* Trace: Lahore → Islamabad

Same map and same hints as the Greedy trace. This time: lowest f = g + h goes first.



Step	Take (lowest f)	g	h	f	Goal?	Frontier now (city: f)
1	Lahore	0	180	180	No	Gujranwala: 200, Sialkot: 240
2	Gujranwala	70	130	200	No	Sialkot: 240, Gujrat: 290, Jhelum: 310
3	Sialkot	40	200	240	No	Gujrat: 250 ↓, Jhelum: 310
4	Gujrat	90	160	250	No	Jhelum: 280 ↓
5	Jhelum	180	100	280	No	Islamabad: 305
6	**Islamabad**	**305**	**0**	**305**	**YES — goal!**	—

Step 2. Gujranwala (f=200) looked best, but its neighbours cost more: Gujrat 290, Jhelum 310. Sialkot (240) is now the lowest — A* goes back!

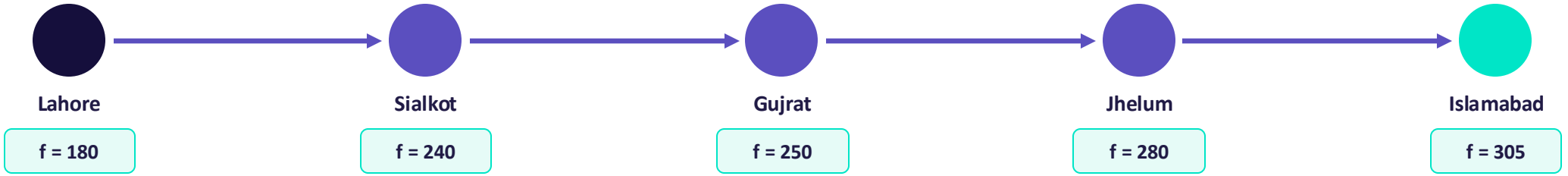
Step 3. Sialkot finds a cheaper way to Gujrat: g = 90, not 130. Gujrat is UPDATED to f = 250 (↓).

PRIORITY QUEUE by f. Total estimated trip cost decides who is served first.

Result: Lahore → Sialkot → Gujrat → Jhelum → Islamabad = 40 + 50 + 90 + 125 = 305 km. The cheapest.

A* Changed Its Mind — Greedy Never Does

Watch f along the cheapest route: it only goes up, and ends at the true cost.



GREEDY

4 cities checked · 335 km

Fast, but fooled by the hint.

A*

6 cities checked · 305 km

Checks a little more, finds the cheapest.

On a tiny map A* looks about as busy as blind search. On a country-sized map, the hint lets A* skip huge areas that UCS would explore.

GUARANTEE #1 · AIMA SECTION 3.5.2

Admissible Heuristic: A Hint That Never Exaggerates

The rule that makes A trustworthy.*

ADMISSIBLE MEANS

h(n) ≤ true remaining cost

The hint may be too low (optimistic), but never too high. It never overestimates.

REAL LIFE

An honest friend says “maybe 100 km” when the road is really 130 km. A friend who says “300 km” when it is 130 km exaggerates — and you might skip a perfectly good route.

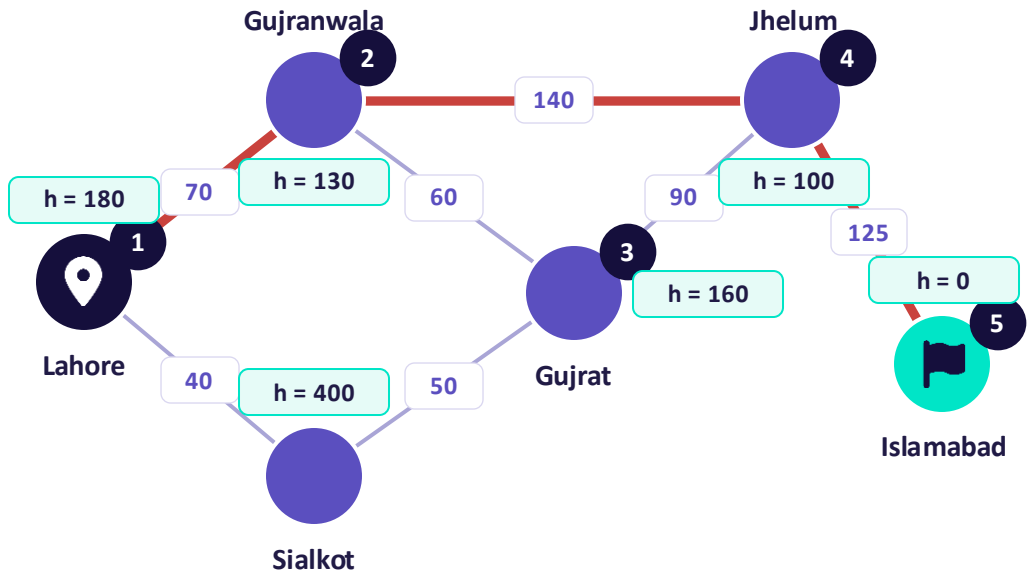
Straight-line distance is always admissible: a road is never shorter than a straight line.

City	Our hint h	True remaining cost	h ≤ true?
Lahore	180	305	✓ yes
Gujranwala	130	265	✓ yes
Sialkot	200	265	✓ yes
Gujrat	160	215	✓ yes
Jhelum	100	125	✓ yes
Islamabad	0	0	✓ yes

The big promise: For A* tree search, an admissible h guarantees optimality. For graph search without reopening explored nodes, h must be consistent.

When the Hint Lies, A* Can Miss the Best Route

Experiment: make Sialkot's hint 400 instead of 200. The true cost from Sialkot is only 265.



Step	Take (lowest f)	g	h	f	Frontier now (city: f)
1	Lahore	0	180	180	Gujranwala: 200, Sialkot: 440
2	Gujranwala	70	130	200	Gujrat: 290, Jhelum: 310, Sialkot: 440
3	Gujrat	130	160	290	Jhelum: 310, Sialkot: 440
4	Jhelum	210	100	310	Islamabad: 335, Sialkot: 440
5	Islamabad	335	0	335	— goal —

Result: A* returned 335 km — the same costly route as Greedy.

Why? Sialkot's f = 440 stayed above the goal's f = 335, so A* did not expand Sialkot before selecting the goal. The cheap route was hidden behind an overestimating hint.

Lesson: An overestimating h can break A*'s optimality guarantee.

Consistency: The Hint Must Not Jump Around

A slightly stronger rule than admissible — and our map follows it.

CONSISTENT MEANS

$$h(n) \leq \text{cost}(n \to n') + h(n')$$

In words: after you drive one road, the hint cannot drop by more than the cost of that road.

WHY IT MATTERS

f never goes down along a path. Our trace: 180 → 240 → 250 → 280 → 305.

So the first time A* takes a city, it has already found the cheapest way to that city.

Every consistent hint is admissible. Straight-line distance is both.

Road	Road cost	Hint changes by	Change ≤ cost?
Lahore – Gujranwala	70	50	✓ yes
Lahore – Sialkot	40	20	✓ yes
Gujranwala – Gujrat	60	30	✓ yes
Gujranwala – Jhelum	140	30	✓ yes
Sialkot – Gujrat	50	40	✓ yes
Gujrat – Jhelum	90	60	✓ yes
Jhelum – Islamabad	125	100	✓ yes

All 7 roads pass: our hints are consistent.

SIDE BY SIDE

UCS vs. Greedy vs. A*

Same search recipe — a different "pick next" rule.

	UCS (blind)	Greedy Best-First	A*
Picks next…	Lowest g (cost so far)	Lowest h (hint)	Lowest g + h
Uses	Real cost only	Hint only	Real cost AND hint
Cheapest route?	**Yes**	**No**	**Yes — admissible (tree); consistent (graph, no reopening)**
Speed	May expand many nodes before reaching the goal	Fast — can be fooled	Usually much faster than UCS with a good hint
Memory	Large	Large in the worst case	Large — its main weakness
Real-life picture	Cheapest-fare finder	Walking toward a tower	Taxi: meter + driver's estimate
Best when…	There is no hint	A quick "good enough" answer will do	You need the cheapest route and have a good hint

Based on Russell & Norvig, AIMA (3rd ed.), Sections 3.4–3.5

IN THE REAL WORLD

Where A* Is Used

A and its relatives are among the most widely used search methods in practice.*



Route planning

Finding the best road: real distance so far (g) plus straight-line distance to the destination (h).



Game pathfinding

Characters find a way around walls and obstacles to reach a target on the game map.



Warehouse & delivery robots

Robots plan a cheap path between shelves, avoiding blocked aisles.



Puzzles & planning

Solving sliding puzzles and planning sequences of actions with a good heuristic.

AVOID THESE TRAPS

Watch Out! Common Mistakes

Four things that cost marks in exams.

Greedy is not “cheapest”.

It is fast, but it can return a costlier route (335 km vs 305 km on our map).

Stop when the goal is TAKEN, not when first seen.

In the Multan activity, Multan appears with f = 300 while Faisalabad (285) is still lower. Keep going until the goal reaches the front.

Always update to the cheaper g.

Gujrat first had g = 130, then Sialkot offered g = 90. Replace the old entry — do not keep both.

For optimality, use an admissible hint.

An overestimate can break A*'s guarantee; graph search without reopening nodes requires consistency.

CHECK YOUR UNDERSTANDING · 6 QUESTIONS

Quick Quiz

Answer on paper or by raising your hand: A, B or C.

Q1 **Greedy Best-First Search chooses the next city using…**

A) h(n) only B) g(n) only C) g(n) + h(n)

Q2 **What is h(goal) — the hint at the goal city itself?**

A) 0 B) 1 C) The road cost

Q3 **A city has g = 60 and h = 90. What is f for A*?**

A) 30 B) 90 C) 150

Q4 **The true remaining cost is 300 km. Which hint is NOT admissible?**

A) 250 B) 300 C) 400

Q5 **If h(n) = 0 for every city, A* behaves like…**

A) DFS B) UCS C) Greedy

Q6 **True or False: Greedy Best-First always finds the cheapest route.**

A) True B) False

QUIZ ANSWERS

Answers & Why

Reveal after the class has answered.

Q1 *Greedy chooses using…*

A) h(n) only — Greedy looks only at the hint, not at the cost already paid.

Q2 *What is h(goal)?*

A) 0 — At the goal there is nothing left to travel.

Q3 *g = 60, h = 90. f = ?*

C) 150 — f = g + h = 60 + 90.

Q4 *True cost 300. Not admissible?*

C) 400 — It is higher than the true cost, so it exaggerates.

Q5 *If h = 0 everywhere…*

B) UCS — With no hint, f = g, which is exactly what UCS uses.

Q6 *Greedy always cheapest?*

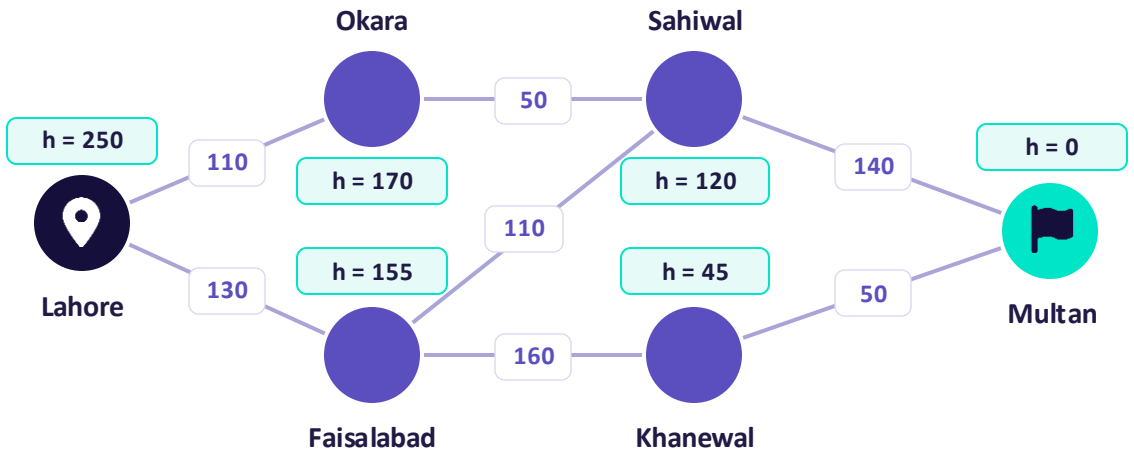
B) False — It is fast but not optimal (335 km vs 305 km).

TRY IT YOURSELF · PAIRS · ~20 MIN · PAPER EXERCISE

Greedy vs. A*: Lahore → Multan

Roads in km (simplified). On a tie, take the cities in alphabetical order.

1. Trace GREEDY with Hint A (smallest h first). Write the order the cities are taken.
2. Trace A* with Hint A. In each step write g, h and f, and the frontier.
3. Fill the table: path, cities checked and total km for both. Which is cheaper?
4. Admissible? Compare Hint A with the true remaining costs in the table.
5. Bonus (Hint C): change Okara's h from 170 to 250. Re-trace A*. What changes, and why?



Hint A is shown on the map.

Cost	Lahore	Okara	Faisalabad	Sahiwal	Khanewal	Multan
Hint A	250	170	155	120	45	0
True cost	300	190	210	140	50	0

Run	Path found	Cities checked	Total km
Greedy			
A*			

ANSWER KEY · REVEAL AFTER THE ACTIVITY

Greedy 340 km vs. A* 300 km

Same map, same hint — A saves 40 km.*

A* TRACE

Step	Take (lowest f)	g	h	f	Goal?	Frontier now (city: f)
1	Lahore	0	250	250	No	Okara: 280, Faisalabad: 285
2	Okara	110	170	280	No	Sahiwal: 280, Faisalabad: 285
3	Sahiwal	160	120	280	No	Faisalabad: 285, Multan: 300
4	Faisalabad	130	155	285	No	Multan: 300, Khanewal: 335
5	**Multan**	**300**	**0**	**300**	**YES — goal!**	—

A*: Lahore → Okara → Sahiwal → Multan = 110 + 50 + 140 = 300 km. Five cities taken. Notice step 3: Multan joins with f = 300, but Faisalabad (285) is lower — so A* continues.

Greedy: Lahore → Faisalabad → Khanewal → Multan = 130 + 160 + 50 = 340 km. Four cities. It never looked at Okara or Sahiwal.

Admissible? Yes ✓
Every h is at most the true cost: 250 ≤ 300, 170 ≤ 190, 155 ≤ 210, 120 ≤ 140, 45 ≤ 50, 0 ≤ 0.

Bonus — Hint C (Okara = 250): 340 km ✗
Okara now has f = 110 + 250 = 360, so A* prefers Faisalabad and Khanewal and returns 340 km. 250 is more than Okara's true cost (190) — an overestimate.

WRAPPING UP

Key Takeaways

If you remember only six things from today:

1. Informed search uses a hint h(n): a cheap guess of the remaining cost. h(goal) = 0.

2. Greedy Best-First uses f(n) = h(n). Fast — but it ignores the cost already paid, so it can be fooled.

3. A* uses f(n) = g(n) + h(n): cost so far plus estimated cost to go. Lowest f goes first.

4. Admissible = never overestimates. This guarantees optimal A* tree search; graph search without reopening nodes needs consistency.

5. Consistent hint = never drops by more than the road cost. Then f never decreases along a path.

6. A better hint means less searching. With h = 0, A* is just UCS. A*'s weakness is memory.

BEFORE NEXT LECTURE

Recommended Reading

PRIMARY TEXT

Russell, S. & Norvig, P. (2010). Artificial Intelligence: A Modern Approach (3rd ed.). Chapter 3, Section 3.5 (Informed Search Strategies: 3.5.1 Greedy best-first search, 3.5.2 A*) and Section 3.6 (Heuristic Functions).

ALSO USEFUL (OPTIONAL)

Text book: Luger, G.F. & Stubblefield, W.A. (2009). AI Algorithms, Data Structures, and Idioms in Prolog, Lisp, and Java — chapters on heuristic search.

Reference book: Norvig, P. — Paradigms of AI Programming (PAIP), search chapters.